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Effective Components of Collaborative Care for Depression in Primary Care
An Individual Participant Data Meta-Analysis

Hannah Schillok, MSc; Jochen Gensichen, MD, MA, MPH; Maria Panagioti, PhD; Jane Gunn, PhD, MBBS; Lukas Junker, MSc; Karoline Lukaschek, PhD; Caroline Jung-Sievers, MD; Philipp Sterner, MSc; Lukas Kaupe, MD; Tobias Dreischulte, PhD, MSc, MPharm; Mohammed K. Ali, MD, MSc, MBA; Enric Aragonès, MD, PhD; David B. Bekelman, MD, MPH; Birgit Herbeck Belnap, PhD; Robert M. Carney, PhD; Lydia A. Chwastiak, MD, MPH; Peter A. Coventry, PhD; Karina W. Davidson, PhD; Maria L. Ekstrand, PhD; Alison Flehr, PhD; Susan Fletcher, PhD; Lars P. Hölzel, PhD; Klaas Huijbregts, PhD; Viswanathan Mohan, MD, PhD, DSc; Vikram Patel, MB, BS, PhD; David A. Richards, PhD; Bruce L. Rollman, MD, MPH, MBA; Chris Salisbury, MD, MSc; Gregory E. Simon, MD, MPH; Krishnamachari Srinivasan, MBBS, MD, DPM; Jürgen Unützer, MD, MPH, MA; Christina M. van der Feltz-Cornelis, MD, PhD; Kenneth B. Wells, MD, MPH; Thomas Zimmermann, PhD; Markus Bühner, PhD; for the POKAL Group

IMPORTANCE Collaborative care is a multicomponent intervention for patients with chronic disease in primary care. Previous meta-analyses have proven the effectiveness of collaborative care for depression; however, individual participant data (IPD) are needed to identify which components of the intervention are the principal drivers of this effect.

OBJECTIVE To assess which components of collaborative care are the biggest drivers of its effectiveness in reducing symptoms of depression in primary care.

DATA SOURCES Data were obtained from MEDLINE, Embase, Cochrane Library, PubMed, and PsycInfo as well as references of relevant systematic reviews. Searches were conducted in December 2023, and eligible data were collected until March 14, 2024.

STUDY SELECTION Two reviewers assessed for eligibility. Randomized clinical trials comparing the effect of collaborative care and usual care among adult patients with depression in primary care were included.

DATA EXTRACTION AND SYNTHESIS The study was conducted according to the IPD guidance of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting guideline. IPD were collected for demographic characteristics and depression outcomes measured at baseline and follow-ups from the authors of all eligible trials. Using IPD, linear mixed models with random nested effects were calculated.

MAIN OUTCOMES AND MEASURES Continuous measure of depression severity was assessed via validated self-report instruments at 4 to 6 months and was standardized using the instrument's cutoff value for mild depression.

RESULTS A total of 35 datasets with 38 comparisons were analyzed (N = 20 046 participants [57.3% of all eligible, with minimal differences in baseline characteristics compared with nonretrieved data]; 13 709 [68.4%] female; mean [SD] age, 50.8 [16.5] years). A significant interaction effect with the largest effect size was found between the depression outcome and the collaborative care component therapeutic treatment strategy (−0.07; $P < .001$). This indicates that this component, including its key elements manual-based psychotherapy and family involvement, was the most effective component of the intervention. Significant interactions were found for all other components, but with smaller effect sizes.

CONCLUSIONS AND RELEVANCE Components of collaborative care most associated with improved effectiveness in reducing depressive symptoms were identified. To optimize treatment effectiveness and resource allocation, a therapeutic treatment strategy, such as manual-based psychotherapy or family integration, may be prioritized when implementing a collaborative care intervention.

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Multimedia

Supplemental content

Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: The POKAL Group investigators appear in Supplement 2.

Corresponding Author: Hannah Schillok, MSc, Institute of General Practice and Family Medicine, LMU University Hospital, LMU Munich, Nußbaumstraße 5, 80336 Munich, Germany (hannah.schillok@med.uni-muenchen.de).

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Depression is one of the leading contributors to global health-related burdens, affecting individuals across all ages, sexes, and settings.[1,2] Primary care serves as the first contact point and key hub for outpatient depression treatment in many countries,[3,4] including the US,[5-7] Germany,[3,8] and the UK, where up to 90% of mental health issues are managed.[9,10] However, patients often do not receive optimal care,[11-13] highlighting the need for improvements in quality and patient-centered, proactive treatment.

A promising approach to achieve this is collaborative care,[14] a multicomponent complex intervention typically implemented in primary care for depression. The concept involves a case manager who works with the primary care physician, acting as a link between patients and health care professionals integrating relevant care services and disciplines.[15,16] Collaborative care has been shown to be significantly more effective for depression than usual care,[17] and it is part of several national guidelines[18-20] as well as recommendations by the World Psychiatric Association[21] and the World Health Organization.[22] However, its implementation remains rare, primarily due to limited resources and unclear understanding of its components.[23,24]

Robust insights are therefore needed to identify which components of collaborative care actually drive its effectiveness. While acknowledging that collaborative care must be tailored to its specific setting,[24] identifying essential components and understanding their mechanisms of action remain critical to facilitate effective adoption in practice.

We used random-effects models with individual participant data (IPD) for improved precision to investigate the components driving the effectiveness of collaborative care for depression in adult patients in primary care settings 4 to 6 months after randomization. Previous systematic reviews have assessed components of collaborative care,[14,25-27] but they lacked a comprehensive framework to assess the intervention systematically, are somewhat outdated, and did not integrate IPD, which we used to improve the precision of our estimations.[28]

Key Points

Question What are the most effective components of collaborative care for adult patients with depressive symptoms in a primary care setting?

Findings This meta-analysis with individual participant data from 20 046 patients showed the largest significant effect size for the therapeutic treatment strategy component, with manual-based psychotherapy and involvement of family as key elements.

Meaning Collaborative care is effective for treating depression in primary care. Implementation should consider therapeutic treatment strategies, including manual-based psychotherapy and involvement of family and friends, as these elements can offer the greatest potential for improving depression and significantly impact the intervention's success.

Methods

The protocol of this IPD meta-analysis was registered in PROSPERO (identifier: CRD42024543787). However, the principal component analysis (PCA) and subsequent analyses represent a deviation from the protocol arising from exploratory analyses. The findings are reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guideline for IPD.[29] Secondary outcomes and subgroup analyses, requiring additional methodological considerations, will be addressed in future work. Owing to the retrospective nature of the study and the use of irreversibly deidentified data, the Ethics Committee of the Medical Faculty at LMU Munich deemed that institutional review board approval and obtaining informed consent were not required.

Data Sources

Searches were conducted in December 2023 in 5 databases (MEDLINE, Embase, Cochrane Library, PubMed, and PsycInfo) using the following concepts: depression, collaborative care, and randomized controlled trial (RCT) (eTable 1 in Supplement 1). We also reviewed reference lists of related systematic reviews and contacted authors of eligible trials to identify additional studies (published or ongoing). The collection of eligible data was ongoing until May 2024.

Eligibility Criteria

Two reviewers assessed for study eligibility. We found studies to be eligible for inclusion if they fulfilled the following criteria: (1) participants were adults (aged ≥18 years) with depression, mixed anxiety and mood disorder, or subthreshold symptoms thereof; (2) participants were treated or cotreated in a primary care setting; (3) a collaborative care intervention (defined as a multicomponent intervention, at minimum including a multiprofessional approach, and enhanced interprofessional communication) was received; (4) treatment was compared with usual care or enhanced usual care; (5) a continuous depression score was an outcome; and (6) the study design was an RCT or cluster RCT.

All eligible studies assessed depressive symptoms with a validated score, namely the Patient Health Questionnaire-9,[30] the Hopkins Symptom Checklist,[31] the Hamilton Depression Rating Scale,[32] the Beck Depression Inventory II,[33] the Center for Epidemiologic Studies Depression Scale,[34] the Clinical Interview Schedule-Revised,[35] or the Patient-Reported Outcomes Measurement Information System-Depression.[36] We standardized the individual depression outcomes within each trial using the score-specific cutoff for mild depression and the pooled standard deviation of the study population at baseline. An overview table with all applied cutoffs, ranges, and references can be found in eTable 2 in Supplement 1. Our outcome of interest was the assessment at 4 to 6 months of follow-up.

Systematic Assessment of the Collaborative Care Interventions' Components

For data extraction, we used a framework of collaborative care that assesses the interventions' components and their intensity. The framework is adapted from a widely accepted model of collaborative care[15] that consists of the following components: (1) multiprofessional approach to patient care (involvement of a primary care physician and at least another health

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professional, eg, a nurse, psychologist, psychiatrist, or pharmacist); (2) structured management plan (evidence-based guidelines and protocols, including pharmacological and nonpharmacological interventions); (3) scheduled patient follow-ups (organized follow-up appointments for interventions, treatment adherence, and symptom monitoring); and (4) enhanced interprofessional communication (mechanisms for communication among professionals, eg, team meetings, case conferences). We added the fifth component, increased patient and family activation (strategies to empower patients and their families, including coping strategies, relapse prevention plans, involving family in care, and considering community or culture), to reflect recent developments in primary care trials and health policy that emphasize patient and carer involvement and empowerment.[37-41] We extracted all components as sum scores of dichotomous items, with all independent covariates treated as continuous variables. A detailed description of how the collaborative care intensity framework was applied, including the operationalization of each component, can be found in eTables 3 and 4 in Supplement 1.

We contacted the authors of all eligible trials via email and asked for the IPD of the following variables: treatment group, age, sex, and baseline and follow-up depression scores. Data were cleaned, were converted to a uniform reporting format, and had depression outcomes standardized. Before merging into a single dataset, we calculated initial separate analyses of the depression outcome to ensure consistency and completeness with reported original data. Missing data were retained as missing. We supplemented the IPD with information on the intervention's content derived from the full texts using a standardized Excel sheet (Microsoft Corp). The process of data extraction was piloted to ensure feasibility and afterward conducted by 2 independent researchers. Initial interrater reliability was assessed by calculating the Cohen κ for every intervention component item (average, 0.66). The respective contingency table can be found in eTable 5 in Supplement 1. Afterward, conflicts were noted, discussed, and resolved by a third reviewer if necessary. We compared data available to us with data unavailable on study population, outcome, intervention intensity, and common effect moderators to check for significant differences. Two independent reviewers checked the quality of the included trials, applying the Cochrane risk-of-bias 2 (RoB 2) tool (eTable 6 in Supplement 1).[42] Further quality checks included depression severity baseline imbalance and missing data (eFigure 1 in Supplement 1).

Statistical Analysis

We conducted a 1-stage meta-analysis using linear mixed models (LMMs) with 1 random intercept for study and an additional random effect for treatment nested within study. We chose this approach to account for between-study heterogeneity in treatment that was otherwise not captured by the fixed effects via a random slope.

Our approach is in line with recommendations by methodology literature.[43] The model we designed contains homoscedastic error variance between studies, as no indication for heteroscedasticity was found while checking graphically (eFigure 2 in Supplement 1).

First, we ran an exploratory LMM with IPD outcome data to empirically test the previously described operationalization of the received collaborative care intensity used for data extraction. The dependent variable was the individual depression outcome after 4 to 6 months, while the independent variables were the 5 collaborative care components based on the received intervention according to the modified framework, controlled for the participants' age, sex, and depression severity at baseline. We thus ran an LMM combining study-level and patient-level covariates. Details on this LMM analysis can be found in eTables 7 and 8 in Supplement 1.

This exploratory allocation led to suppression effects, which we found to stem from pronounced collinearity particularly between 2 components (Pearson correlation coefficient, 0.61; eTable 9 in Supplement 1), making interpretation difficult. We thus decided to conduct a PCA for dimensionality reduction.[44] PCA assigns items to components based on their empirical correlations, ensuring that highly correlated items are grouped together.[45]

We used parallel analysis to determine the number of components for PCA, resulting in 4 components. Details on the PCA and the parallel analysis are described in eTable 10 in Supplement 1. The resulting allocation of items was then applied to the LMM.

For analyses, we used R packages psych[46] and ImerTest[47] (R Foundation) as well as Stata version 17.0 (StataCorp LLC)[48] statistical software. Two-tailed $P < .05$ was considered statistically significant.

Before analysis, we descriptively assessed the course of depression, stratified by treatment group over time using all individual depression outcomes across all available assessment points, ranging from month 1 to month 48 after randomization, via the R package ggplot2.[49]

Results

As shown in **Figure 1**, we found a total of 74 RCTs (35 395 participants) with 78 comparisons to be eligible for our IPD meta-analysis. Based on the recommended[50] funnel plot for these studies, we could not detect evidence for significant asymmetry (Egger test [SE], −0.09 [0.08]; $P = .27$; eFigure 3 in Supplement 1). We collected data from 35 trials (20 273 participants [57.3% of total]) including 38 comparisons. A complete list of included studies can be found in eTable 11 in Supplement 1. In total, 227 individuals (1.1%) were excluded from the analyses due to their age (<18 years) or because only the depression subgroup of the trial was eligible for our purpose, resulting in 20 046 unique cases.

Baseline Characteristics and Comparison Between Available and Unavailable Data

Eighteen studies stemmed from the US, 11 from Europe, 3 from India, 2 from Australia, and 1 from Canada. Most participants (13 709 [68.4%]) were female, and the mean (SD) age was 50.8 (16.5) years (range, 18-95 years). We did not encounter any important issues related to data accuracy or completeness while checking the IPD. Details on the characteristics of all studies as well as their duration can be found in eTables 12 and 13 in

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Supplement 1. We compared available and unavailable studies based on population, design, intervention content, and outcomes, informed by previous reviews, which identified these factors as potential effect modifiers in collaborative care.[14,27] As shown in **Table 1**, we did not find any significant differences except for the intervention content (represented as collaborative care intensity framework). A higher percentage of more intense trials provided IPD.

Course of Depression Over Time

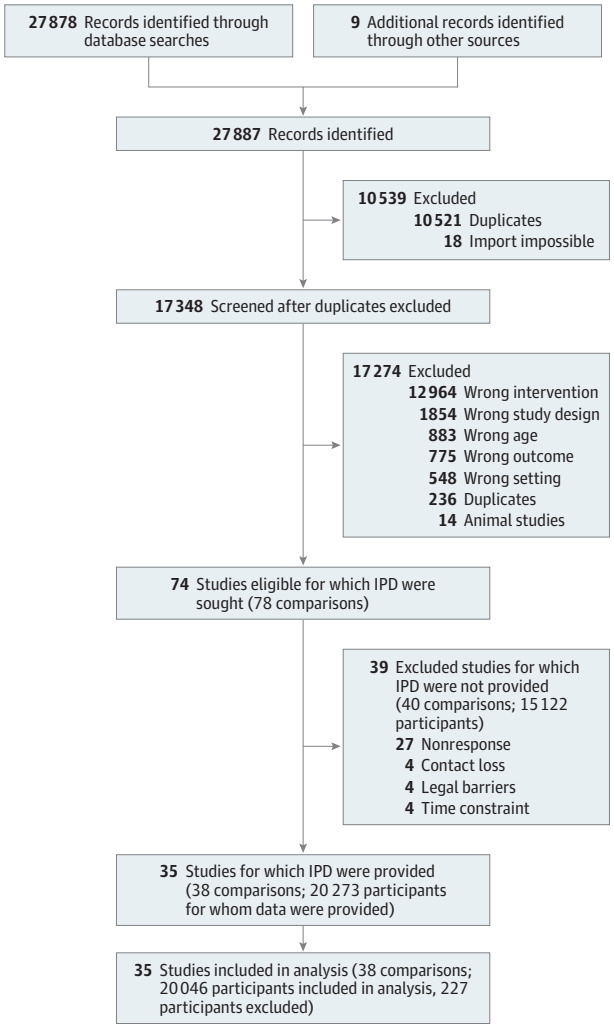
In our data, the treatment group receiving collaborative care showed reduced depression severity compared with the usual care group at each assessment time, persisting for the first 24 months (**Figure 2**). A corresponding forest plot for the overall effect of collaborative care at 4 to 6 months confirmed this pattern, with collaborative care having a small significant effect over usual care (standardized mean difference, −0.20 [95% CI, −0.26 to −0.15]; I^2 = 58.4%; eFigure 4 in Supplement 1). Information on data availability at each assessment time can be found in eTable 14 in Supplement 1.

Association Between Depression Outcome and Intervention Components

To resolve collinearity observed with the intervention components, we rearranged their extracted items via PCA, which yielded the following 4 components, also described in more detail in **Table 2** and eTable 15 in Supplement 1: (1) patient-centered care (individualized care respecting patient preferences, needs, and values); (2) therapeutic treatment strategy (structured therapeutic approaches and support for effective depression management); (3) measurement-based care (systematic data-driven monitoring and treatment plan adjustments); and (4) integrated mental health care (comprehensive linkage for primary mental health care).

When applying the LMM with random nested effects, a statistically significant association was found between the therapeutic treatment strategy component and the standardized depression outcome (−0.07; P < .001; **Table 3**), indicating that this component may be particularly effective in reducing depression severity. The primary items that loaded most heavily on this component were manual-based psychotherapy and involvement of family or friends (eTable 10 in Supplement 1). We also saw significant effects for the other components but with smaller effect sizes of −0.04 each, indicating a decrease in depression severity when the component is more intensively implemented.

Figure 1. Flowchart Illustrating the Study Selection Process



IPD indicates individual participant data.

Discussion

This IPD meta-analysis found that the derived collaborative care component labeled *therapeutic treatment strategy*, including its main items manual-based psychotherapy and involvement of family or friends, was most effective for reducing depression severity in primary care. While the reported individual effect of this key component may appear modest, it is straightforward to implement and has the potential to benefit a large number of patients in the primary care setting.

Furthermore, we found evidence that the other derived components contributed to the effectiveness of collaborative care to some degree as well, albeit with smaller effects on reducing depression severity. Considering these findings and potentially positive interactions between these components, we believe that all components should be implemented to some extent. However, our findings clearly indicate that the therapeutic treatment strategy warrants greater intensity and resource allocation to improve patient depression outcomes.

Comparison With Existing Systematic Reviews

The effectiveness of collaborative care for depression in primary care has been well established in the literature.[17,25] The question of the most effective components of collaborative care has been addressed already, but findings have remained inconclusive so far.[51,52] Previous studies have used various methodological approaches, including descriptive or narrative syntheses of systematic reviews,[53-56] observational studies,[51,57,58]

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Table 1. Comparison of 35 Trials That Provided IPD vs 39 Trials That Did Not Provide IPD

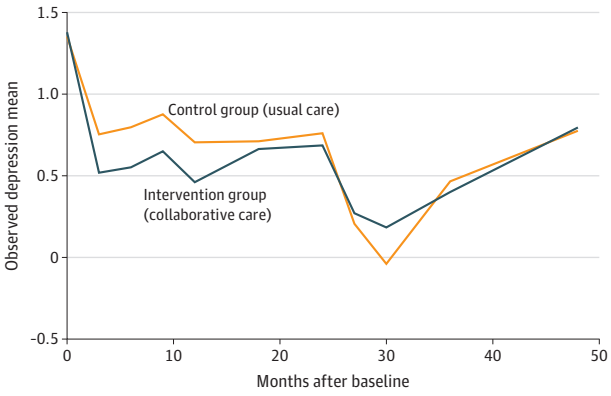
Variable	IPD availability		Statistical test[a]	*P* value
	Available (n = 35 trials)	Unavailable (n = 39 trials)		
Participants, No.	20 046	15 122		
Based in United States, No. (%)	18 (51.4)	21 (55.3)	$\chi^2_1 = 0.108$	.74
Publication date, year				
Mean (SD)	2010.7 (7.6)	2010.8 (7.9)	$t_{71} = -0.026$	.98
Median (range)	2012.0 (1995-2022)	2010.5 (1995-2022)		
PHQ-9 used as depression screening tool, No. (%)	15 (42.9)	9 (23.7)	$\chi^2_1 = 3.035$	.08
Retention rate, %				
Mean (SD)	82.7 (11.3)	84.5 (8.8)	$t_{72} = -0.749$	.46
Median (range)	85.8 (40.8-96.3)	84.0 (61.1-98.9)		
Population size, No.				
Mean (SD)	572.7 (604.8)	397.9 (368.4)	$t_{55.2711} = 1.476$	.15
Median (range)	329 (65-2486)	345 (74-2365)		
Female participants per trial, %				
Mean (SD)	65.5 (18.0)	70.0 (19.3)	$t_{71} = -1.021$	.31
Median (range)	72.1 (3.4-82.9)	71.6 (3.5-100)		
Mean age of participants, y				
Mean (SD)	50.8 (16.5)	55.5 (12.1)	$t_{71} = -1.137$	.26
Median (range)	48.3 (35.6-77.9)	55.3 (37.4-77.3)		
Physical condition present, No. (%)	11 (31.4)	8 (21.1)	$\chi^2_1 = 1.019$	.31
Intervention duration, mo				
Mean (SD)	8.6 (4.4)	8.9 (6.6)	$t_{71} = -0.201$	.84
Median (range)	7 (3-24)	6 (2-36)		
Collaborative care intensity framework score				
Mean (SD)	10.8 (2.4)	9.0 (2.5)	$t_{71} = 3.213$	.002
Median (range)	11 (5-16)	9 (2-14)		
Self-reported significant outcome, No. (%)[b]	29 (82.9)	29 (76.3)	$\chi^2_1 = 0.478$	.49

Abbreviations: IPD, individual participant data; PHQ-9, Patient Health Questionnaire–9.

[a] Group comparisons were conducted using *t* tests or χ^2 tests, with prior variance homogeneity assessed via Levene tests.

[b] Responses were dichotomized.

Figure 2. Depression Course Over Time Shown With Individual Participant Data



The depression score is standardized, with the y-axis indicating the number of standard deviations by which the score is above the cutoff for mild depression.

and metaregressions with aggregated data.[25-27,59] These studies as well as similar studies on the chronic care model[60] have provided valuable groundwork despite facing certain methodological challenges, such as limited causal inference[53] or reliance on preselected components.[14,25-27] Our study contributes to this ongoing effort by systematically assessing the components of collaborative care for depression and incorporating IPD for improved model estimations.

A primary finding of our analysis was that the therapeutic treatment strategy was the most effective component for reducing depression severity, particularly its main items manual-based psychotherapy and involvement of family or friends. This aligns with a prior meta-analysis,[27] which highlighted psychological interventions as the only significant predictor of improved depression outcomes in collaborative care among 10 preselected predictors. Similar findings are reported by a systematic review.[55] Our results support and extend these findings by also identifying the critical role of family and friend involvement—a previously understudied component in collaborative care.[61] This aspect, which may range from psychoeducation to active participation in patient care, has been shown to bring tangible clinical benefits to patients' mental health outcomes in the literature[62] and also appears to enhance the effectiveness of collaborative care interventions.

Implications for Clinicians, Policymakers, and Researchers

Considering that collaborative care components can be implemented with varying intensities, we provide valuable in-

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sights into where resources could be more intensively allocated to improve depression outcomes. The most influential component—therapeutic treatment strategy, with its key items of manual-based psychotherapy and involvement of family or friends—should be prioritized for implementation.

While our findings do not suggest that other components are completely ineffective or should be disregarded, identifying the most influential components is crucial for improving complex interventions, such as collaborative care. Practitioners and policymakers should ensure this key component is consistently included in future intervention designs to optimize effectiveness. Additionally, these findings offer an initial basis for engaging health insurers to evaluate coverage decisions. Funding critical components may enhance the impact of collaborative care on depression outcomes and support sustainable implementation in routine practice.

In terms of research implications, it may be helpful to further untangle collaborative care, separating active ingredients from delivery approaches to assess their relative importance. This may require experimental approaches that manipulate specific intervention components, such as factorial experiments (eg, 2 × 2, 3 × 3),[63,64] allowing for testing of combinations and interactions of core components, or adaptive approaches, including sequential multiple assignment randomized trials[65] and the multiphase optimization strategy.[66] Coupled with data on real-world implementation,[67] which has already been shown to be particularly important in the collaborative care field,[68] such an approach may confirm our results and inform the development and broader adoption of structured depression care.

Table 2. Components of Collaborative Care After Principal Component Analysis

Component	Items	Dichotomized outcome
Patient-centered care	Consideration of patient preference	No vs yes
	Involvement of community or cultural background	No vs yes
	Goal setting	No vs yes
	Automatization process for follow-up	No vs yes
	Form of monitoring	By phone vs in person
	Supervising mental health specialist	No vs yes
	Regularly scheduled patient reviews	No vs yes
	Written documentation or team meeting	No vs yes
Therapeutic treatment strategy	Manual-based psychotherapy	No vs yes
	Involvement of family or friends	No vs yes
	Organization of treatment team	Centralized vs locally
	Counseling (≤8 sessions)[a]	No vs yes
	Routine follow-up with case manager every 2-4 wk	No vs yes
	Disease-related coping strategies	No vs yes
Measurement-based care	Intensive follow-up with case manager at least every 2 wk	No vs yes
	Ad hoc emergency communication	No vs yes
	Shared medical record	No vs yes
Integrated mental health care	Case manager with mental health background	No vs yes
	Cotreatment by mental health specialist	No vs yes
	Relapse prevention plan	No vs yes

[a] In addition to potential medication therapy.

Strengths and Limitations

We believe this study is the most rigorous methodological examination of the most effective collaborative care components to date and the first attempt to synthesize these data also incorporating IPD.[27] We exploratorily used an a priori, conceptually developed model of collaborative care, which we then contrasted with an empirically driven, data-derived model offering a complementary perspective.

Our sole focus on primary care adds a new, clear perspective. To address potential biases, we tested for funnel plot asymmetry, finding no significant publication bias, and used comprehensive literature searches (ie, major databases, snowballing, author requests) with strict, piloted inclusion criteria to mitigate study selection bias. Using LMMs incorporating random nested effects, alongside controlling for key covariates using IPD, we were able to improve our model precision.[28] Although robust, our findings should be interpreted cautiously, as complete elimination of all biases could not be ensured, with mainly 4 notable limitations remaining.

First, we accessed 57% of IPD across requested RCTs, which is below the recommended recruitment target (ie, 80% of published data)[50] but comparable to similar meta-analyses.[69,70] While baseline characteristics showed minimal differences between studies with and without IPD, unmeasured factors may still affect our findings, highlighting the need for improved data sharing. Second, while we assessed the interventions systematically, the complexity and inconsistent reporting of collaborative care[17,71,72] posed some challenges in recording details uniformly. Integrating IPD with study-level data was essential for our research, although it reduced variance and impacted statistical power. Third, while our framework captured key components of collaborative care, it did not account for variables such as the physician-patient relationship[73] or the trustworthiness between collaborative care stakeholders.[74-76] These factors, shaped by trust, familiarity, and clear role definitions, may significantly influence intervention outcomes and merit further exploration. Additionally, while we included age, sex, and baseline depression severity as covariates in our analyses, other potentially important factors, such as employment status,[77] were not consistently available across studies. Future research with more consistent data availability could provide a more comprehensive understanding of the factors influencing the effectiveness of collaborative care interventions. Fourth, our data did not allow us to fully disentangle the cluster components combining specific intervention content with intensity. Despite this limitation, our analysis provides critical insights into the active ingredients of collaborative care, which have remained unclear until now.[17,59]

Conclusions

To our knowledge, this IPD meta-analysis provides the most rigorous and conclusive insights into the most effective

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Table 3. Linear Mixed Model With Nested Random Effects Examining the Effect of 4 Collaborative Care Components According to Principal Component Analysis on the Depression Outcome Adjusted for Patient-Level Age, Sex, and Baseline Depression[a]

Variable	Estimate	SE	*df*	*t* Value	*P* value
Intercept	−0.02	0.06	37.43	−0.32	.75
Patient-centered care	−0.04	0.02	18.88	−2.33	.03
Therapeutic treatment	−0.07	0.02	13.63	−4.08	<.001
Measurement-based care	−0.04	0.02	37.24	−2.45	.02
Integrated mental health care	−0.04	0.02	20.35	−2.85	.01
Age	0.00	0.00	12 908.93	2.52	.01
Female	0.01	0.02	13 502.21	0.34	.73
Depression baseline score	0.47	0.01	13 144.15	55.88	<.001

[a] Higher levels of depressive symptoms were found among women and with increasing age.

components of collaborative care for depression in primary care that should be considered by implementers of this complex intervention. A strong focus should thus lie on the derived component therapeutic treatment strategy, including patient and family involvement (eg, in the form of psychotherapy, psychological short interventions, involvement of family members, and coping skills training).

ARTICLE INFORMATION

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Author Affiliations: Institute of General Practice and Family Medicine, LMU University Hospital, LMU Munich, Munich, Germany (Schillok, Gensichen, Lukaschek, Kaupe, Dreischulte); Graduate Program, POKAL—Predictors and Outcomes in Primary Care Depression Care, Munich, Germany (Schillok, Gensichen, Junker, Lukaschek, Jung-Sievers, Kaupe, Dreischulte, Bühner); German Center for Mental Health, Munich/Augsburg, Germany (Gensichen, Bühner); National Institute of Health and Care Research (NIHR) School for Primary Care Research, NIHR Greater Manchester Patient Safety Research Collaborations, Division of Population Health, Health Services Research and Primary Care, School of Health Sciences, Faculty of Biology, Medicine and Health, University of Manchester, Manchester, United Kingdom (Panagioti); National Institute of Health and Care Research Greater Manchester Patient Safety Translational Research Centre, Division of Population Health, Health Services Research and Primary Care, University of Manchester, Manchester, United Kingdom (Panagioti); Department of General Practice and Primary Care, Melbourne Medical School, The University of Melbourne, Melbourne, Victoria, Australia (Gunn, Flehr, Fletcher); Department of Psychology, LMU Munich, Munich, Germany (Junker, Sterner, Bühner); Institute of Medical Information Processing, Biometry and Epidemiology, Faculty of Medicine, LMU Munich, Munich, Germany (Jung-Sievers); Pettenkofer School of Public Health Munich, Munich, Germany (Jung-Sievers); German Center for Mental Health, Bochum, Germany (Sterner); Emory Global Diabetes Research Center, Woodruff Health Sciences Center and Emory University, Atlanta, Georgia (Ali); Department of Family and Preventive Medicine, School of Medicine, Emory University, Atlanta, Georgia (Ali); Institut Universitari d'Investigació en Atenció Primària Jordi Gol, Barcelona, Spain (Aragonès); Department of Medicine, Eastern Colorado Health Care System, Denver (Bekelman); Division of General Internal Medicine, Department of Medicine, Anschutz Medical Campus, University of Colorado School of Medicine, Aurora (Bekelman); Department of Veterans Affairs, Eastern Colorado Health Care System, Denver (Bekelman); Center for Behavioral Health, Media, and Technology, University of Pittsburgh School of Medicine, Pittsburgh, Pennsylvania (Herbeck Belnap, Rollman); Department of Psychiatry, Washington University School of Medicine in St Louis, St Louis, Missouri (Carney); Department of Global Health, University of Washington, Seattle (Chwastiak); Department of Psychiatry and Behavioral Sciences, University of Washington, Seattle (Chwastiak, Unützer); Mental Health and Addiction Research Group, Department of Health Sciences, University of York, York, United Kingdom (Coventry, van der Feltz-Cornelis); York Environmental Sustainability Institute, University of York, York, United Kingdom (Coventry); Northwell, New Hyde Park, New York (Davidson); Division of Mental Health and Neurosciences, St John's Research Institute, Bengaluru, India (Ekstrand, Srinivasan); Division of Prevention Sciences, Department of Medicine, University of California, San Francisco (Ekstrand); Health Services Research, Oberberg Clinic Group, Berlin, Germany (Hölzel); Department of Psychiatry and Psychotherapy, University Medical Center Mainz, Mainz, Germany (Hölzel); Department of Psychiatry and Psychotherapy, Medical Center–University of Freiburg, Faculty of Medicine, University of Freiburg, Freiburg, Germany (Hölzel); Scelta (part of GGNet Mental Health Care), Apeldoorn, Warnsveld, the Netherlands (Huijbregts); Department of Diabetology, Madras Diabetes Research Foundation and Dr Mohan's Diabetes Specialities Centre, Chennai, India (Mohan); Department of Global Health and Social Medicine, Harvard Medical School, Boston, Massachusetts (Patel); Department of Health and Caring Sciences, Western Norway University of Applied Sciences, Bergen, Norway (Richards); Centre for Academic Primary Care, Bristol Medical School, University of Bristol, Bristol, United Kingdom (Salisbury); Kaiser Permanente Washington Health Research Institute, Seattle (Simon); Department of Psychiatry, St John's Medical College, Bengaluru, India (Srinivasan); Department of Psychiatry and Biobehavioral Sciences, David Geffen School of Medicine, University of California, Los Angeles (Wells); Jane and Terry Semel Institute for Neuroscience and Human Behavior, University of California, Los Angeles (Wells); Department of General Practice and Primary Care, Centre for Psychosocial Medicine, University Medical Centre Hamburg-Eppendorf, Hamburg, Germany (Zimmermann).

Author Contributions: Ms Schillok had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Schillok, Gensichen, Panagioti, Jung-Sievers, Kaupe, Herbeck Belnap, Carney, Chwastiak, Coventry, Davidson, Ekstrand, Mohan, Srinivasan, van der Feltz-Cornelis, Wells, Bühner.
Acquisition, analysis, or interpretation of data: Schillok, Gensichen, Panagioti, Gunn, Junker, Lukaschek, Sterner, Dreischulte, Ali, Aragonès, Bekelman, Chwastiak, Davidson, Ekstrand, Flehr, Fletcher, Hölzel, Huijbregts, Patel, Richards, Rollman, Salisbury, Simon, Srinivasan, Unützer, van der Feltz-Cornelis, Zimmermann, Bühner.
Drafting of the manuscript: Schillok, Unützer.
Critical review of the manuscript for important intellectual content: Gensichen, Panagioti, Gunn, Junker, Lukaschek, Jung-Sievers, Sterner, Kaupe, Dreischulte, Ali, Aragonès, Bekelman, Herbeck Belnap, Carney, Chwastiak, Coventry, Davidson, Ekstrand, Flehr, Fletcher, Hölzel, Huijbregts, Mohan, Patel, Richards, Rollman, Salisbury, Simon, Srinivasan, Unützer, van der Feltz-Cornelis, Wells, Zimmermann, Bühner.
Statistical analysis: Schillok, Junker, Sterner, Bühner.
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Supervision: Gensichen, Panagioti, Lukaschek, Jung-Sievers, Herbeck Belnap, Davidson, Mohan, Zimmermann, Bühner.

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